

AI Automation, Regime Type, and Crossed Fragility

Abstract

How does AI-driven automation affect the stability of democracies versus autocracies? We develop a formal model showing that the *trajectory* of automation, not just its magnitude, determines which regime type is more fragile. We compare two stylized trajectories borrowed from the O-Ring automation framework (Gans and Goldfarb 2026): rapid displacement, where workers lose jobs immediately, and threshold automation, where AI initially complements labor and later substitutes it. We find a crossed fragility result: democracies are stable under rapid displacement but fragile under threshold automation, while autocracies exhibit the opposite pattern. The mechanism is the coordination capacity of affected workers. Rapid shocks generate common knowledge of grievance, enabling organized opposition that democracies channel into compensation but that degrades autocratic repression. autocratic repression but channels into democratic compensation. Threshold shocks produce strategic uncertainty that prevents coordination, paralyzing democratic response while preserving repressive effectiveness. We show that comprehensive social insurance achieves the same institutional resilience as repressive capacity, with strictly superior welfare outcomes. The model generates a five-type regime typology with testable predictions about which countries are most vulnerable to which automation trajectory.

1 Introduction

The United States and China lead the world in generative artificial intelligence. One is a democracy; the other is an autocracy. The economic debate over AI has focused on employment, wages, and productivity. The political question has received less attention: how does AI-driven automation affect regime stability, and does the answer depend on regime type? Historically, major technological transformations reshaped the class structures that produced both democratization and authoritarianism (Boix 2003; Acemoglu and Robinson 2006). AI-driven automation is likely to be no exception, and the stakes are amplified by the geopolitical competition between democratic and autocratic adopters.

Recent economic research shows that automation can follow qualitatively different trajectories depending on the structure of production (Autor, Levy, and Murnane 2003; Acemoglu and Restrepo 2020). When tasks within a job are complementary, partial automation initially raises productivity; substitution occurs only after a threshold is crossed (Gans and Goldfarb 2026). In some sectors, automation displaces workers immediately. In others, it complements them first and replaces them later. The political implications of this distinction have not been explored.

The regime stability literature treats economic shocks as uniform events defined by magnitude, not trajectory. Acemoglu and Robinson (2006) show that transitory shocks open windows for regime transitions, but treat transitoriness as a binary property. Svoblik (2012) identifies two problems of authoritarian rule, power-sharing and mass control; our model formalizes the second. Przeworski (2005, 2006) models democracy as a self-enforcing equilibrium that is static in income: it does not address what happens when income is disrupted, or how the shape of disruption matters. The empirical record suggests that democracies are fragile under economic crises while autocracies prove resilient (Przeworski et al. 2000), but no formal model explains this pattern or identifies the conditions under which it reverses.

The intuition is direct. If automation displaces workers immediately, democratic governments face visible protests and respond with compensation; autocratic governments face organized opposition that strains their security forces. If automation initially augments workers but replaces them after a threshold, democratic governments see no crisis and build no protection. When the shock arrives, it is too late. Autocratic governments, whose standing security apparatus does not require a crisis to exist, contain the disoriented workers without difficulty. The same technology produces opposite political outcomes depending on its trajectory.

This paper formalizes that logic. We build a two-period model in the Open Economy Politics tradition (Lake 2009), taking the economics of automation as given (from Gans and Goldfarb 2026) and deriving the political-institutional implications.

First, we establish a crossed fragility pattern (Proposition 3). For autocracies with moderate repressive capacity, democracies are stable under rapid displacement but unstable under threshold automation; autocracies exhibit the reverse. The mechanism is the coordination capacity of affected workers. Rapid shocks generate common knowledge of grievance (Chwe 2001), enabling collective action that democracies channel into compensation but that degrades autocratic repression (Kuran 1991; Edmond 2013). Threshold shocks produce strategic uncertainty that prevents coordination, paralyzing democratic response while preserving repressive effectiveness.

Second, we show that the welfare state functions as institutional insurance against delayed shocks (Propositions 7-8). Democracies with comprehensive social protection survive threshold automation through the same logic as autocracies with standing repressive capacity: both represent permanent institutional capacity that does not require crisis-triggered activation. The welfare state achieves this resilience while compensating workers; repression achieves it while leaving them to bear the full cost. The welfare gap between a surviving democracy and a surviving autocracy equals exactly the repressive threshold, $\bar{\kappa}$.

Przeworski and Wallerstein (1982) modeled the class compromise sustaining democratic capitalism: workers accept private profit in exchange for social protection (see also Moene and Wallerstein 2001). Our model formalizes a specific implication. When this compromise is institutionalized as a welfare state, it insures the regime itself against technological disruption. The authoritarian substitute, repression as regime insurance (Desai, Olofsgard, and Yousef 2009), achieves equivalent stabilization at the cost of leaving citizens uncompensated.

The paper proceeds as follows. Section 2 presents the model. Section 3 derives the main

results. Section 4 develops the welfare state extension and the complete regime typology. Section 5 discusses implications and limitations. Appendices A and B provide game-theoretic microfoundations for the two behavioral assumptions.

2 The Model

2.1 Environment

We consider a polity over two periods $t \in \{1, 2\}$ with a group of workers E exposed to AI-driven automation. The remainder of the economy is captured by the parameter $W > w_E$, representing total per capita output. The polity operates under one of two regime types $i \in \{D, A\}$: democracy or autocracy.

Automation trajectories. Following Gans and Goldfarb (2026), we distinguish two stylized trajectories of automation. Under *rapid displacement* ($j = r$), automation causes immediate job loss: $\ell_1 = L$, $\ell_2 = 0$, where $L > 0$ is the magnitude of the shock. Under *threshold automation* ($j = \tau$), automation initially complements labor and only later becomes substitutive: $\ell_1 = 0$, $\ell_2 = L$. These trajectories are exogenous to the political model — they represent the economic premise imported from the automation literature.

Preferences. In each period, each member of E chooses between M (moderate — accept the status quo) and P (protest — pursue regime change). Payoffs depend on the regime type.

Under democracy:

$$\begin{aligned} u_E(M \mid D, t) &= w_E - \ell_t(1 - \varphi_t) \\ u_E(P \mid D, t) &= \theta W - k \end{aligned}$$

Under autocracy:

$$\begin{aligned} u_E(M \mid A, t) &= w_E - \ell_t \\ u_E(P \mid A, t) &= \theta W - k - \kappa_0 \cdot \eta_j \end{aligned}$$

where $\varphi_t \in [0, 1]$ is the compensatory capacity of the democracy, $\kappa_0 > 0$ is the autocracy's standing repressive capacity, $\eta_j \in (0, 1]$ is repressive effectiveness (indexed by trajectory j), $\theta \in (0, 1)$ is the share of total output captured by E under regime change, and $k > 0$ is the cost of transition (violence, uncertainty, capital destruction).

The key structural asymmetry: democracy stabilizes by making M more attractive (compensation reduces losses), while autocracy stabilizes by making P more costly (repression raises the price of revolt).

2.2 Assumptions

Define the auxiliary variable $\Delta \equiv \theta W - k - w_E$. We say a regime is *stable* when E prefers M to P ; the formal definition appears in Section 3.

A1 (Stability without automation). $\Delta < 0$. Without automation, E prefers M under both regimes. The regime is inherently stable.

A2 (Automation can destabilize). $\Delta + L > 0$, equivalently $L > |\Delta|$. The automation shock is large enough to potentially trigger regime change.

A3 (Democratic compensation is reactive).

$$\varphi_t = \begin{cases} \bar{\varphi} & \text{if } \ell_t > 0 \\ 0 & \text{if } \ell_t = 0 \end{cases}$$

where $\bar{\varphi} \equiv 1 - |\Delta|/L \in (0, 1)$ is the *compensatory threshold*.

Democratic compensation requires political legitimacy — legislative authorization and fiscal mobilization — that emerges only under observable crisis. This pattern is historically robust: major social insurance programs are created after crises (the New Deal after the Great Depression, the welfare state after World War II), not before. Appendix A provides a formal microfoundation: in a voting game, the non-exposed majority vetoes taxation when no crisis is observed (the threat of regime change is not credible) and approves it when losses are visible (the threat becomes credible).¹

A4 (Repressive effectiveness depends on shock type).

$$\eta_R \in (0, 1) \quad (\text{rapid: degraded repression})$$

$$\eta_T = 1 \quad (\text{threshold: full repression})$$

Rapid shocks are visible and shared, creating near-common knowledge of grievance among affected workers. This facilitates coordination: organized opposition degrades repressive effectiveness through security force defection, international pressure, and the moral legitimacy of the cause. Threshold shocks are sudden and disorienting, producing strategic uncertainty that impedes coordination: dispersed dissent is easily contained by standing repressive apparatus. Appendix B provides a formal microfoundation via global games (Morris and Shin 2003): under rapid shocks, high signal precision yields an equilibrium with high participation in collective action; under threshold shocks, noisy private signals destroy coordination.²

A5 (Capacity is a stock). Capacity built in $t = 1$ persists to $t = 2$. No new capacity can be built in $t = 2$.

Example. Let $\Delta = -0.4$, $L = 1$, $\eta_R = 0.5$, $\kappa_0 = 0.8$. Then $\bar{\varphi} = 0.6$, $\bar{\kappa} = 0.6$, $\bar{\kappa}/\eta_R = 1.2$. Since $\kappa_0 = 0.8 \in [0.6, 1.2)$, this autocracy is in the *crossed fragility* range. Consider the four scenarios:

¹The result holds for any investment rule $\varphi(\ell)$ satisfying $\varphi(0) = 0$ and $\varphi(L) \geq \bar{\varphi}$. The specific functional form of A3 is not required.

²The mechanism is informational, not preferential: E desires revolt equally under both shock types ($u_E(P)$ is the same). What differs is the capacity to coordinate, determined by the information structure of the shock.

- *Democracy, rapid* ($\ell_1 = 1$): Crisis is visible. Compensation activates: $\varphi_1 = 0.6$. Workers' payoff under M : $w_E - 1 \cdot (1 - 0.6) = w_E - 0.4$. Under P : $w_E + \Delta = w_E - 0.4$. Workers are indifferent — marginally **stable**.
- *Democracy, threshold* ($\ell_1 = 0, \ell_2 = 1$): No crisis in $t = 1$, no compensation built. In $t = 2$: $\varphi_2 = 0$. Workers' payoff under M : $w_E - 1$. Under P : $w_E - 0.4$. Workers prefer P — **unstable**.
- *Autocracy, rapid* ($\ell_1 = 1$): Workers are organized (visible crisis). Effective repression: $\kappa_0 \cdot \eta_R = 0.8 \times 0.5 = 0.4 < \bar{\kappa} = 0.6$. Repression insufficient — **unstable**.
- *Autocracy, threshold* ($\ell_2 = 1$): Workers are disoriented (sudden shock). Effective repression: $\kappa_0 \cdot \eta_T = 0.8 \times 1.0 = 0.8 \geq 0.6$. Repression sufficient — **stable**.

Same shock magnitude, opposite outcomes by regime and trajectory. These parameters are used throughout Figures 1 and the corollaries.

3 Results

3.1 Stability conditions

Definition. Regime i is *stable under trajectory* j if $u_E(M \mid i, t) \geq u_E(P \mid i, t)$ for all t .

Democracy. When $\ell_t = 0$: stable by A1. When $\ell_t = L$ and $\varphi_t = \bar{\varphi}$: $u_E(M) = w_E + \Delta = u_E(P)$, stable. When $\ell_t = L$ and $\varphi_t = 0$: stability requires $\Delta + L \leq 0$, contradicting A2. *Unstable*.

The *compensatory threshold* is $\bar{\varphi} = 1 - |\Delta|/L$.

Autocracy. When $\ell_t = 0$: stable. When $\ell_t = L$: stable iff $\kappa_0 \cdot \eta_j \geq \Delta + L \equiv \bar{\kappa}$.

The *repressive threshold* is $\bar{\kappa} = L - |\Delta|$.

Since $\eta_R < 1$: the autocracy needs $\kappa_0 \geq \bar{\kappa}/\eta_R$ to survive rapid shocks but only $\kappa_0 \geq \bar{\kappa}$ to survive threshold shocks. Rapid displacement is *harder* for autocracies to contain.

3.2 Main propositions

Proposition 1 (Democratic fragility pattern). *Under A1-A3 and A5: (a) democracy is stable under rapid displacement; (b) democracy is unstable under threshold automation.*

Proof. The proof has two parts, one for each trajectory.

- Rapid displacement.* The shock hits in period 1: $\ell_1 = L > 0$. Because the crisis is observable, democratic compensation activates (A3): $\varphi_1 = \bar{\varphi}$. By construction of $\bar{\varphi}$, this exactly offsets the destabilizing effect of the shock — workers are indifferent between M and P , so the regime is (marginally) stable. In period 2, there is no shock ($\ell_2 = 0$), so stability holds trivially by A1.
- Threshold automation.* Period 1 is calm ($\ell_1 = 0$), so no compensation is triggered: $\varphi_1 = 0$ (A3). The shock arrives in period 2 ($\ell_2 = L$), but by A5 no new capacity can

be built. With $\varphi_2 = 0$, stability would require $\Delta + L \leq 0$, which contradicts A2. The democracy is unstable: workers prefer P to an uncompensated shock. ■

Proposition 2 (Autocratic fragility pattern). *Under A1, A2, A4: (a) autocracy is stable under rapid displacement iff $\kappa_0 \geq \bar{\kappa}/\eta_R$; (b) autocracy is stable under threshold automation iff $\kappa_0 \geq \bar{\kappa}$.*

Proof. (a) *Rapid.* The shock hits in period 1: $\ell_1 = L$. Workers coordinate (the crisis is visible and shared), so repressive effectiveness is $\eta_R < 1$. The effective repressive capacity is $\kappa_0 \cdot \eta_R$. Stability requires $\kappa_0 \cdot \eta_R \geq \bar{\kappa}$, i.e., $\kappa_0 \geq \bar{\kappa}/\eta_R$. Period 2 is calm. (b) *Threshold.* Period 1 is calm. The shock arrives in period 2: $\ell_2 = L$. Workers are disoriented (the shock is sudden), so $\eta_T = 1$ — repression operates at full capacity. Stability requires $\kappa_0 \geq \bar{\kappa}$. ■

Proposition 3 (Crossed fragility). *Under A1-A5, if $\kappa_0 \in [\bar{\kappa}, \bar{\kappa}/\eta_R)$: (a) under rapid displacement, democracy is stable and autocracy is unstable; (b) under threshold automation, democracy is unstable and autocracy is stable.*

Proof. First, note that the interval $[\bar{\kappa}, \bar{\kappa}/\eta_R)$ is nonempty because $\eta_R < 1$ implies $\bar{\kappa}/\eta_R > \bar{\kappa}$.

- (a) *Rapid displacement.* Democracy is stable by P1(a) — the observable crisis triggers compensation. The autocracy, however, faces organized opposition: $\kappa_0 < \bar{\kappa}/\eta_R$ (by the upper bound of the interval), so repression is insufficient by P2(a). Democracy survives; autocracy does not.
- (b) *Threshold automation.* Democracy is unstable by P1(b) — no crisis was observed, no compensation was built. The autocracy’s standing repression, however, operates at full effectiveness against disoriented workers: $\kappa_0 \geq \bar{\kappa}$ (by the lower bound of the interval), so stability holds by P2(b). Autocracy survives; democracy does not. ■

Proposition 3 is the paper’s central result. Each regime has *differential institutional resilience* for one type of shock: democracies process observable shocks better (coordination channels into compensation); autocracies process delayed shocks better (repression contains disoriented opposition).

Proposition 4 (Welfare cost of autocratic stability). *Conditional on regime survival, E ’s welfare is strictly higher under democracy:*

$$u_E(M \mid D, \text{stable}) - u_E(M \mid A, \text{stable}) = \bar{\kappa} > 0$$

Proof. Democracy stable (rapid, $t = 1$): $u_E = w_E + \Delta$. Autocracy stable (threshold, $t = 2$): $u_E = w_E - L$. Difference: $\Delta + L = \bar{\kappa} > 0$. ■

Autocratic stability is “stability without welfare.” The regime survives but affected workers absorb the full automation shock. The welfare gap is exactly the repressive threshold $\bar{\kappa}$.

Proposition 5 (Threshold of thresholds). *Define $L^* \equiv \kappa_0 + |\Delta|$. For $L > L^*$, autocracy is unstable under threshold automation.*

Proof. Stability requires $\kappa_0 \geq \bar{\kappa} = L - |\Delta|$. $L > \kappa_0 + |\Delta| \Rightarrow \bar{\kappa} > \kappa_0$. ■

Sufficiently large shocks overwhelm even standing repressive capacity. The “threshold of thresholds” is the point where automation exceeds repression.

Proposition 6 (Width of the crossed interval). *The set of κ_0 values generating the crossed pattern has width $\bar{\kappa}(1 - \eta_R)/\eta_R$, which is increasing in $\bar{\kappa}$, decreasing in η_R , and zero when $\eta_R = 1$.*

Proof. $|\bar{\kappa}/\eta_R - \bar{\kappa}| = \bar{\kappa}(1 - \eta_R)/\eta_R$. Derivatives are immediate. ■

The parameter η_R governs everything. It captures how vulnerable autocratic repression is to organized opposition. When $\eta_R = 1$ (repression is equally effective regardless of coordination), the crossed pattern vanishes; the model collapses to one where trajectory does not matter for autocracies. The crossed fragility result is a consequence of $\eta_R < 1$ — the empirically grounded assumption that repression degrades against organized, visible, legitimate opposition.

Corollary 1 (Autocratic typology).

Type	Condition	Rapid	Threshold
Democracy	(by A3)	Stable	Unstable
Weak autocracy	$\kappa_0 < \bar{\kappa}$	Unstable	Unstable
Moderate autocracy	$\bar{\kappa} \leq \kappa_0 < \bar{\kappa}/\eta_R$	Unstable	Stable
Strong autocracy	$\kappa_0 \geq \bar{\kappa}/\eta_R$	Stable	Stable

4 Extension: The Welfare State as Institutional Insurance

The baseline model (Propositions 1-6) assumes that democracies have no standing compensatory capacity ($\varphi_0 = 0$). But some democracies maintain comprehensive social insurance that does not require crisis activation. We now ask: does standing compensatory capacity protect democracies against threshold automation, in the same way that standing repressive capacity protects autocracies?

Generalized A3. Allow standing compensatory capacity $\varphi_0 \geq 0$:

$$\varphi_t = \max(\varphi_0, \bar{\varphi} \cdot \mathbf{1}[\ell_t > 0])$$

Proposition 7 (Welfare state as insurance against delayed shocks). *(a) Democracy is stable under threshold automation iff $\varphi_0 \geq \bar{\varphi}$. (b) For $\varphi_0 < \bar{\varphi}$, Proposition 1 holds unchanged. (c) For $\varphi_0 \geq \bar{\varphi}$, democracy is stable under both trajectories.*

Proof. (a) Threshold, $t = 2$: $\ell_2 = L$, $\varphi_2 = \varphi_0$. Stable iff $\varphi_0 \geq \bar{\varphi}$. (b) $\varphi_0 < \bar{\varphi}$: threshold $t = 2$ has $\varphi_2 = \varphi_0 < \bar{\varphi}$, unstable; rapid $t = 1$ has $\varphi_1 = \bar{\varphi}$, stable. Identical to P1. (c) $\varphi_0 \geq \bar{\varphi}$: in any period with $\ell_t = L$, $\varphi_t \geq \varphi_0 \geq \bar{\varphi}$, stable. ■

The parameter φ_0 is the democratic analogue of κ_0 . Both represent *permanent institutional capacity* functioning as insurance against delayed shocks: κ_0 through repression, φ_0 through social protection. Democracies with comprehensive welfare states ($\varphi_0 \geq \bar{\varphi}$) are functionally equivalent to strong autocracies ($\kappa_0 \geq \bar{\kappa}/\eta_R$) in terms of resilience — but with strictly superior welfare outcomes.

Proposition 8 (Functional equivalence with welfare gap). *Both strong democracies ($\varphi_0 \geq \bar{\varphi}$) and strong autocracies ($\kappa_0 \geq \bar{\kappa}/\eta_R$) are stable under both trajectories. But under threshold automation, E 's welfare differs:*

$$u_E(M \mid D\text{-strong}) - u_E(M \mid A\text{-strong}) \geq \bar{\kappa} > 0$$

Proof. Strong democracy, threshold $t = 2$: $u_E = w_E - L(1 - \varphi_0) \geq w_E - L(1 - \bar{\varphi}) = w_E + \Delta$. Strong autocracy, threshold $t = 2$: $u_E = w_E - L$. Difference $\geq \bar{\kappa} > 0$ by A2. ■

This result formalizes a precise claim: the welfare state and repressive apparatus are functionally equivalent as *stabilization* mechanisms but not as *welfare* mechanisms. A strong democracy stabilizes *and* compensates. A strong autocracy stabilizes but leaves workers to bear the full cost of automation. The normative argument for comprehensive social insurance is that it purchases the same institutional resilience as repression — without the human cost.

This connects to the class compromise tradition. Przeworski and Wallerstein (1982) modeled the bargain sustaining democratic capitalism: workers consent to private profit in exchange for social protection (see also Moene and Wallerstein 2001 on the demand for social insurance under inequality). Our P7-P8 formalize a specific implication: when this compromise is institutionalized as a standing welfare state, it insures the regime itself — not just individual workers — against technological disruption. The authoritarian alternative, welfare spending as a substitute for (or complement to) repression to prevent insurrection (Desai, Olofsgard, and Yousef 2009), achieves stabilization but at the cost of leaving citizens uncompensated.

4.1 Why democracies cannot recover in $t = 2$: endogenous fiscal fragility

A natural objection: under threshold automation, the crisis *is* observable in $t = 2$ ($\ell_2 = L$). Why can't the democracy build emergency compensation then? Assumption A5 rules this out by fiat. We now relax A5 and show that the result survives under an endogenous fiscal constraint.

A5' (Emergency fiscal constraint). In $t = 2$, the democracy can attempt to build emergency compensatory capacity φ_2^{em} at cost $p \cdot c(\varphi_2^{em})$, where $p \geq 1$ is the *crisis premium* (reflecting fiscal stress, political polarization, and borrowing costs under duress) and $c(\cdot)$ is the normal cost of building capacity. Emergency spending is subject to a fiscal constraint: $p \cdot c(\varphi_2^{em}) \leq F$, where $F > 0$ is available fiscal capacity.

Proposition 9 (Endogenous fiscal fragility). *Under A5' replacing A5, define $p^* \equiv F/c(\bar{\varphi})$. Democracy is unstable under threshold automation whenever $p \geq p^*$.*

Proof. Under threshold, $t = 2$: $\ell_2 = L$, $\varphi_1 = 0$ (by A3). Survival requires $\varphi_2^{em} \geq \bar{\varphi}$, costing $p \cdot c(\bar{\varphi})$. This is affordable iff $p \cdot c(\bar{\varphi}) \leq F$, i.e., $p \leq p^*$. For $p \geq p^*$: emergency compensation is unaffordable. P1(b) holds. ■

Three forces push p above p^* during a threshold crisis:

1. *Fiscal erosion.* The automation shock shrinks the tax base: workers who lose income pay less tax. The fiscal capacity F itself declines with ℓ_2 .
2. *Political paralysis.* Workers moving toward P makes legislative bargaining over emergency spending harder. The non-exposed majority (who must approve taxation, per Appendix A) faces a more polarized environment.
3. *Credibility collapse.* Emergency programs announced during crisis are less credible than established institutions: implementation lags, administrative capacity is absent, and beneficiaries cannot verify promises.

The asymmetry with autocracy is structural: κ_0 is a pre-built stock, not a crisis-mobilized flow. The repressive apparatus does not require emergency fiscal authorization — its cost was paid *ex ante* through continuous maintenance. Compensation, by contrast, must be fiscally mobilized *during* the crisis, precisely when mobilization is hardest.

Corollary 3 (Vicious cycle of democratic fragility). *Under threshold automation with endogenous fiscal constraint, weak democracies ($\varphi_0 < \bar{\varphi}$) face a vicious cycle: (i) no preventive investment in $t = 1$ (A3); (ii) crisis in $t = 2$ erodes fiscal capacity ($F \downarrow$) and raises political costs ($p \uparrow$); (iii) emergency response becomes unaffordable ($p > p^*$); (iv) instability. Preventive investment ($\varphi_0 \geq \bar{\varphi}$, P7) breaks this cycle by building capacity before fiscal conditions deteriorate.*

This result strengthens the normative case for the welfare state. The choice is not merely between compensating workers (humane) and repressing them (effective). It is between investing *before* the crisis, when fiscal conditions permit, and attempting to invest *during* the crisis, when they may not. A5 is the limiting case ($p \rightarrow \infty$); the endogenous version preserves the result whenever crisis conditions are severe enough to make emergency response prohibitively expensive.

Corollary 2 (Complete regime typology).

Type	Condition	Rapid	Threshold
Strong democracy	$\varphi_0 \geq \bar{\varphi}$	Stable	Stable
Weak democracy	$\varphi_0 < \bar{\varphi}$	Stable	Unstable
Weak autocracy	$\kappa_0 < \bar{\kappa}$	Unstable	Unstable
Moderate autocracy	$\bar{\kappa} \leq \kappa_0 < \bar{\kappa}/\eta_R$	Unstable	Stable
Strong autocracy	$\kappa_0 \geq \bar{\kappa}/\eta_R$	Stable	Stable

The crossed fragility pattern (Proposition 3) applies to the comparison between *weak democracies* and *moderate autocracies*. Figure 1 illustrates the parametric regions.

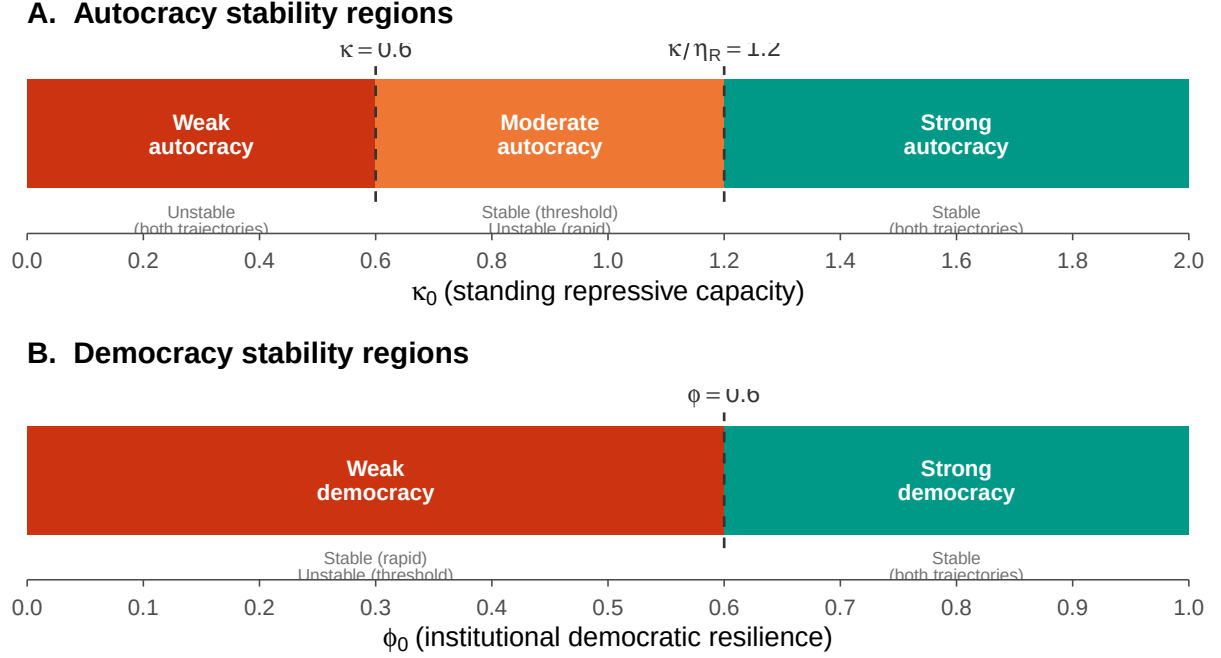


Figure 1: Stability regions by regime type. Panel A: autocracy stability as a function of standing repressive capacity κ_0 . Panel B: democracy stability as a function of standing compensatory capacity φ_0 . Parameters: $L = 1$, $|\Delta| = 0.4$, $\eta_R = 0.5$.

Empirical mapping.

Theoretical type	Candidate examples
Strong democracy ($\varphi_0 \geq \bar{\varphi}$)	Scandinavia, Germany (comprehensive welfare state)
Weak democracy ($\varphi_0 < \bar{\varphi}$)	United States, United Kingdom (residual welfare state)
Moderate autocracy ($\bar{\kappa} \leq \kappa_0 < \bar{\kappa}/\eta_R$)	China, Russia, Iran
Strong autocracy ($\kappa_0 \geq \bar{\kappa}/\eta_R$)	North Korea, Saudi Arabia
Weak autocracy ($\kappa_0 < \bar{\kappa}$)	Competitive autocracies pre-Arab Spring

5 Discussion

5.1 Two symmetric tragedies

The model reveals a structural irony at the heart of each regime type.

The tragedy of democratic responsiveness. The same property that makes democracies good for citizens — responsiveness to political demands, leading to compensatory policy — makes them fragile to delayed shocks. Without observable distress, there is no political demand; without demand, there is no action. Responsiveness is both the source of democratic strength and the mechanism of democratic fragility.

The tragedy of autocratic repression. The same property that makes autocracies resilient to delayed shocks — standing repressive capacity maintained independently of economic conditions — makes them (i) vulnerable to visible shocks, where organized opposition degrades repressive effectiveness, and (ii) unable to benefit citizens even when stability is preserved, since repression contains discontent without addressing its cause.

5.2 Comparative statics

Parameter	Effect on stability	Interpretation
$L \uparrow$	$\bar{\varphi} \uparrow, \bar{\kappa} \uparrow$	Larger shocks destabilize both
$\eta_R \downarrow$	Crossed interval widens	More autocracies exhibit the crossed pattern
$\theta \uparrow$	$\Delta \uparrow$, both thresholds rise	Regime change more attractive
$k \uparrow$	$\Delta \downarrow$, both thresholds fall	Regime change costlier, more stability
$\kappa_0 \uparrow$	Autocracy more stable	Moves from weak to moderate to strong
$\varphi_0 \uparrow$	Democracy more stable under threshold	Welfare state as insurance

5.3 Transferability

The crossed fragility result generalizes beyond AI automation. For any economic shock with two possible trajectories — observable versus delayed — the model predicts that responsive institutions process observable shocks better and permanent-capacity institutions process delayed shocks better. The coordination of affected populations is the mediating variable: observable shocks facilitate it (benefiting institutions that channel coordination); delayed shocks impede it (benefiting institutions that operate independently of coordination). Applications include climate change (gradual warming versus sudden extreme events), financial crises (slow-building versus sudden crashes), and demographic transitions (gradual aging versus sudden labor force contraction).

5.4 Limitations

The model assumes regime-specific instruments: democracies compensate, autocracies repress. In practice, regimes use mixed strategies — autocracies also redistribute (Knutson

and Rasmussen 2018) and democracies also coerce. An extension showing that instrument specialization emerges endogenously from cost advantages would strengthen the argument.

The parameter η_R is assumed, not estimated. Empirically, η_R should correlate with security force professionalization, ethnic cohesion of the military, international isolation, and surveillance technology. Systematic measurement remains a task for future research.

The two-period structure captures timing but not richer dynamics. A multi-period extension with learning and adjustment would allow for intermediate scenarios (partially gradual shocks) and adaptive institutional responses.

6 Conclusion

We have shown that the trajectory of AI automation — rapid displacement versus threshold substitution — produces systematically different fragility patterns across regime types. The key mechanism is the coordination capacity of affected workers, which has opposite effects on democracies (enables compensation) and autocracies (degrades repression).

Three results stand out. First, the crossed fragility pattern: each regime has comparative advantage in processing one type of shock (Proposition 3). Second, the welfare cost of autocratic resilience: when autocracies survive, their citizens bear the full cost of disruption (Proposition 4). Third, the functional equivalence of social insurance and repressive capacity as institutional insurance, with the former strictly dominating in welfare (Propositions 7-8).

The policy implication is direct. Democracies facing AI automation with potential threshold dynamics — sectors where automation initially complements and later substitutes — should invest in standing social insurance *before* the crisis materializes. This is not merely a humanitarian argument. It is a regime-stability argument: the welfare state is the democratic substitute for the repressive apparatus that autocracies maintain as a matter of course. The difference is that social insurance protects both the regime and its citizens, while repression protects only the regime.

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A Microfoundation of A3 — A Voting Model of Reactive Compensation

A.1 Setup

Two groups within the democracy: E (exposed, mass μ_E) and N (non-exposed, mass $\mu_N > \mu_E$). Within each period, after ℓ_t is revealed: (1) E announces political stance (M or P); (2) N votes on a subsidy rate s_t financing compensation φ_t ; (3) E takes final action.

N 's payoff: $U_N = w_N - s_t c_s - \gamma \cdot \mathbf{1}[P \text{ occurs}]$, where $c_s > 0$ is the per-unit cost of the subsidy and $\gamma \equiv (1 - \alpha)w_N > 0$ is the cost N bears under regime change.

A.2 Equilibrium

When $\ell_t = 0$: By A1, E cannot credibly threaten P (revolt is dominated even without compensation). N 's optimal tax is $s^* = 0$.

When $\ell_t = L$: E 's threat is credible (by A2). N compares pacification ($s = \bar{\varphi}$, payoff $w_N - \bar{\varphi}c_s$) versus instability ($s = 0$, payoff $w_N - \gamma$). N pacifies iff $\gamma \geq \bar{\varphi}c_s$.

Parameter restriction (P-R): $\gamma/c_s \geq \bar{\varphi}$. Under (P-R), the unique subgame-perfect equilibrium yields $\varphi^*(0) = 0$ and $\varphi^*(L) = \bar{\varphi}$, recovering A3 as an equilibrium result.

The microfoundation rests on three primitives: majority rule ($\mu_N > \mu_E$), shock-contingent outside option (automation activates E 's threat), and N 's fear of instability ($\gamma > 0$). The non-exposed majority is the bottleneck: it funds social insurance only under duress.

B Microfoundation of A4 — Coordination as Global Game

B.1 Setup

A continuum of E -members indexed by i , each choosing $a_i \in \{0, 1\}$ (participate in revolt or abstain). Revolt succeeds iff participation $\pi \equiv \int a_i di \geq \bar{\pi} \equiv \kappa_0/\kappa_{\max}$. Each agent observes a private signal $x_i = \omega + \sigma \varepsilon_i$ about shock severity, where signal noise σ depends on the trajectory:

- Rapid ($\sigma_\tau \approx 0$): visible, shared experience. Near-common knowledge.
- Threshold ($\sigma_\tau \gg 0$): sudden, disorienting. Noisy private signals.

Net benefit of successful revolt: $B = \theta W - k - (w_E - L) = \bar{\kappa} > 0$ (by A2). Cost of failed participation: $c > 0$.

B.2 Equilibrium

Rapid shock ($\sigma \rightarrow 0$): The game approaches complete information. The risk-dominant equilibrium features high participation when $B/(B + c) > \bar{\pi}$, which holds for moderately repressive autocracies. Organized opposition.

Threshold shock (σ large): Strategic uncertainty dominates. Each agent's posterior about others' participation is diffuse. The unique Bayesian Nash equilibrium features low participation: $\pi_\tau^* < \bar{\pi}$. Revolt fails. Disorganized dissent.

B.3 Mapping to η

Define $\eta(j) = h(\pi_j^*)$ where h is decreasing, $h(0) = 1$. Under rapid: π^* high $\Rightarrow \eta_R < 1$ (organized opposition degrades repression). Under threshold: π^* low $\Rightarrow \eta_T = 1$ (repression fully effective). This recovers A4 as an equilibrium outcome.

The mechanism is informational: E desires revolt equally under both shocks, but can only coordinate under rapid. Common knowledge enables collective action (Chwe 2001); strategic uncertainty prevents it (Kuran 1991).³

³Przeworski and Limongi (1997) established the empirical precursor: economic development does not cause transitions to democracy, but existing democracies at higher income levels survive longer. Our model provides a mechanism for why economic disruptions differentially affect regime survival depending on both the trajectory of the disruption and the type of regime.